

Synthetic Data Generation of Adverse Driving Conditions for Autonomous Driving Pretraining

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Approval

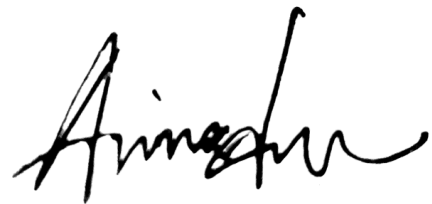
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Ethics Statement

We have cited all the information we have used in our research work. Our knowledge in any of the papers didn't create any bias while reviewing other papers.

Abstract

To improve the safety and reliability of Autonomous Vehicles (AVs) in chaotic traffic environments, we are developing a Synthetic Dataset (SD) for Autonomous Vehicles (AVs) focusing on the difficulty of traffic and human behaviors, in order to enhance the safety and reliability of AVs in chaotic traffic scenarios. The main purpose of our work is to create high quality and diverse datasets with advanced simulation. These datasets enable AVs to be well suited for the common road networks found in South Asia, which are normally unstructured. At present, most perception systems are trained on roads from the western part of the world, which are more structured. This leads to these systems breaking down at every intersection at which rickshaws cross lanes, pedestrians walk between moving vehicles, and rickshaws (which do not exist in any western dataset) are found. If a Nosimon were coming down a street in front of a bus, and there was a pedestrian running down the street in the middle of the street, that's not the kind of thing you would want to have a camera crew go and film. Hence, we created a "data factory" based on a simulation, which was capable of generating any visual class or behaviour edge case that we required, in large volume, without risking anyone's life. We want the autonomy that learns from this data not only to perceive the road accurately, but to understand the social logic of South Indian streets, where the signals are non-autonomous and the flow is not the one in Western cities, and where the street is the one that is often unpredictable, and one that no Western benchmark has tried to seriously understand.

Keywords: Synthetic Data, End to End models, CNN, Diffusion

Acknowledgement

We would like to thank our supervisor Mr Avinandan Banerjee for his expert guidance, patience and mentorship throughout the process of this research. He was engaged from the outset, sitting beside us as we struggled through the early stages to define the problem in the right way, challenging us when we were in need of it, and always reminding us of the harder and also more honest questions. Helpful suggestions he gave had a significant impact in the development of this thesis, especially in the process of producing synthetic data. We are particularly thankful for his helpful feedback which challenged us to think outside the box from a technical standpoint.

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Chapter 1

Introduction

1.1 Background

Driving assistance is becoming crucial in the modern era for covering long distances by road, but achieving semi-autonomous driving functionality comes with caveats as it has not reached the level of safety humans expect from a modern machine. The goal of this paper is to make a leap forward in making semi-autonomous driving a safer feature through synthetically generating adverse road conditions and training it to bridge the safety gap in autonomous driving. Achieving human-level driving capabilities requires bridging this gap in simulated driving datasets. Currently, collecting real-world data to train driving models is prohibitively expensive and logistically dangerous. The cost of physically recreating "edge cases" such as multi-car pileups or pedestrians darting out in heavy fog is a burden that ultimately falls on the end-user. Synthetic data generation via computer simulation offers a revolutionary solution to this problem [2]. Unlike real-world collections, synthetic generation provides researchers with high-level control over generation metrics.

1.2 Rational of the Study

This paper aims to achieve full safe driving on the road. Modern Autonomous Vehicles has reached self-driving capabilities as we expect, but with earlier versions of it having flaws along with. This is due to reasons varying from overwhelming the onboard computers with sensory data to corner cases of unpredictable human behavior. Car crash investigations gave us quite a lot of knowledge on how the model works while driving on road. Car accidents take away thousands of life every year. Autonomous vehicle crashes occur due to reasons such as different car sensors (LiDAR, Radar) oscillating between multiple object classification and responses to different objects, leading to discarded tracking, then to reasons like reaction lag, where the system takes a short time to compute and prevent false positives at the moment of impact. Even reasons such as humans engaging auto-drive mode, giving less attention to driving, and at the moment of a sudden collision they take time to gain consciousness of the moment. Although autonomous vehicles suffer mostly from classification errors, Waymo, a car driving technology company, as mostly solved this issue with their NIEON (Non-Impaired, Eyes always ON) human model and their L4 model, which persistently tracks and classifies objects, allowing them

to outperform even idealized human models through which Waymo reduced crash injuries to about 92%. This paper aims to generally improve the accuracy even further.

This paper also aims to bridge the existing gap in the simulated driving dataset to reach the driving capabilities achieved by humans. This can be done through computer generated simulations, a core necessity in achieving driving safety. The problem with driving on the road and collecting data to train driving models is that it is usually very expensive in nature, and this cost has to be beared by the end user. So, solving this issue can be revolutionary to automated driving assistance, which can be achieved by generating data, synthetically, through computer image generation [4]. Synthetically generated datasets will allow researchers to have a high-level control over the generation metrics, such as driving scenarios in heavy fog, heavy rain, clogged up vision sensors, complex traffic scenes, driving edge cases etcetera. Making a model to work perfectly requires a large portion of a training dataset, and synthetic data generation just happens to be the perfect choice for medium cost with high effectiveness.

Training data volume aside, diversity in training data and quality of data are also crucial in training a model. Problems such as overfitting, which is a neural network learning all training data but losing the ability to generalize, which in concept is a failure of the Bias-Variance trade-off can make model predictions unreliable. Model entropy is the capability of a network models parameter distribution with extremely high randomness or noise, a crucial concept in information theory and statistical learning, referring to the variety of functions a model can fit, which in concept is a deep neural network (DNN) with millions of parameters, that represents a hypothesis space with massive complexity (often discussed in terms of VC-dimension or Rademacher complexity). Problems such as exploding gradient, which is the need for specific activation functions, which in concept is during backpropagation, gradients are calculated via the chain rule, involving repeated multiplication of the weight matrices, can become unstable if a large amount of generated data is provided. To make training efficient, we considered these problems and made synthetic data generation efficient to make model training and prediction effective.

1.3 Problem Statement

The scarcity of sufficient training data, often the unavailability of general driving data, is the gap that this paper aims to reduce. Critical safety events, such as pedestrians crossing the road directly in front of the vehicle, occur most often in countries in the South Asian region, making it easy to collect data for robust model prediction training. There are no training data for 3-wheelers, a common medium of transportation in countries like Bangladesh, India, and Pakistan. Also, the on-road traffic behavior is different from the Western countries. So, generating synthetic data for training driving models, to eradicate the region specificity, will also result in significant accuracy in model performance.



Figure 1.1: Gathering Pedestrians, adapted from <https://en.bonikbarta.com/bangladesh/iuyHDYXARmGfr9lA>.

1.4 Objective

The main goal of our research is to build a strong system for creating synthetic data. This system will simulate traffic conditions that are chaotic and mixed like that of South Asia. To achieve this, we must appeal to a large-scale data factory that can produce scenes on the fly that can look like scenes from the roads of Bangladesh without endangering any camera crew. Today’s roads are autonomous, and they reflect that. That is the starting point, and our study attempts to take a step to address the issue.

We have three concrete objectives in our mind as we study:

Indigenous Asset Creation: They will not reliably detect one on the road if they had never seen a Nosimon or an easy-bike in training. To solve this, we create realistic visualizations of these vehicles, which are not part of nuScenes, BDD100K or any other public benchmark available today.

Adverse Scenario Simulation: Complex driving scenarios with non-lane following of cars will be simulated. The scenarios will be based on the “Drivership” of the local driving community. Variations in vehicles entering and out, and the “creeping” of traffic will be modeled. Unpredictable pedestrian behaviour will also be incorporated such as jaywalking in busy urban environments.

Pretraining & Validation: : We want to show that it helps to pretrain models

for object detection. We will use YOLO architectures for this test. First, we will train them using our synthetic data set. We will then evaluate them using actual video footage of roads in South Asia. Hopefully this will help them to more easily identify and classify local vehicles.

1.5 Methodology in Brief

The methodology constructs a synthetic data factory with real dashcam data and enriches it with images from AI image editing creating rare, dangerous and dangerous scenarios that would not otherwise be possible or feasible to film intentionally. Each generated frame is first created based on a real road photo-graph (including its true geometry, lighting and infrastructure), and then Stable Diffusion XL is tasked with painting an agent or event into a precisely outlined area of the image, using three structural maps generated from the original scene and a text prompt that describes exactly what is supposed to appear, and how surrounding traffic should react. Some vehicle classes, such as rickshaws or auto-rickshaws, that are rarely encountered by the base model are also served by a small number of targeted adapters, trained on just a handful of images per class, yielding agents that are truly local instead of merely approximated. When we need to show the motion, not just one moment, our pipeline will create 2 endpoint frames and give them to a separate video model, which will fill in the motion in between, creating physically believable motions instead of blending pixels. The output may be a single photo or a video clip, and it goes through an automatic verification step to ensure the photo contained the objects the detector was looking for, to accurately boundary the objects' pixel footprint, to ensure that if the objects were in a physical scene, it followed basic physical laws, and to directly write the resulting bounding boxes to the same annotation format that is used for training the detector, without any manual labelling. The final dataset combines these synthetic and real-world data sets: the detector is first trained on the synthetic data to learn the rare concepts and then fine-tuned on real footage, and whenever the pipeline's output is below quality thresholds, the settings are adjusted and the entire generation run again until the output reaches the required quality.



Figure 1.2: Research Methodology

1.6 Scopes and Challenges

The scope of our work is deliberately narrow not because the broader problem is uninteresting, but because doing one thing well matters more than doing everything partially. What we are specifically trying to close is the gap between what standard datasets contain and what South Asian roads actually look like. That focus shapes everything below.

Region-Specific Asset Generation: The most immediate gap is visual. Standard datasets like nuScenes, BDD100K, and their peers simply do not contain auto-rickshaws, battery-powered easy-bikes, or the range of three-wheelers that make up a substantial fraction of Bangladeshi road traffic. Creating accurate visual representations of these vehicles is the foundation everything else rests on.

Unstructured Scenario Simulation: We limit our research to urban, mixed-traffic environments. These areas are known for non-lane-based driving. We focus on generating specific difficult scenarios. Examples include pedestrians walking on the road, sudden cut-ins by small vehicles, and driving through intersections without traffic lights.

Perception Task Focus: TPath planning and full end-to-end driving are deliberately out of scope here. What we are testing is narrower: whether a vision-based detector specifically a YOLO model which does become meaningfully more accurate at recognising local vehicle classes after being exposed to our synthetic data. That is the one question we are set up to answer.

Pretraining Methodology: Our experimental claim is that pretraining on region-specific synthetic data gives the model a head start that carries through into the fine-tuning stage. Because BadODD is small, a model arriving at it with no prior exposure to local classes will struggle to generalise from it. We want to find out how much of that struggle disappears when the synthetic pretraining phase is included.

However, our study faces several technical and logistical limits:

The Reality Gap: The sim-to-real problem is not something we have fully solved even no one has. Even with diffusion-based generation starting from real photographs, there will be artifacts and texture inconsistencies that do not appear in genuine dashcam footage. Whether those imperfections are enough to limit what the detector actually learns from the synthetic data is something we will have to measure rather than assume.

Behavioral Fidelity: For us, arguably the harder limitation is behavioral. South Asian traffic runs on a kind of ongoing negotiation between road users subtle gestures, reading gaps, anticipating what the rickshaw three vehicles ahead is about to do — and none of our template-based scenario generation truly captures that. We can model specific events, but the underlying social logic of the street is much harder to encode.

2D Asset Scarcity: High-quality 2D models for local vehicles are not sold in stores. You cannot buy a model of a damaged "Leguna" or a "Nosimon". We must create these assets from scratch. This takes a lot of manual work. Also, our models might lack the variety of shapes found in the real world.

Validation Data Scarcity: We need real-world data to verify our results. However, annotated datasets from Bangladesh are scarce. We only have small sets like BadODD. This limits our ability to test our results on a large scale.

1.7 Team Overview

Our team of absolutely excellent people are conducting the research on this paper. We split our work in the same manner so that there is no gap left in our study. Rafiul Islam Rafi worked on crash statistics to find real gaps in safe driving. Anirban Roy collected thesis papers and organized them based on categories, along with reviewing papers. Mir Ryan Mahmud Shafin reviewed papers and contributed in literature review. Murchona Anika Swapnil contributed in reviewing papers.

1.8 Key Learnings and Insights

Based on the procurement of paper reviews, the key learning is that standard Western autonomous driving datasets are fundamentally ill-suited for the chaotic, heterogeneous traffic of South Asia due to a critical "Reality Gap" in both visual assets and behavioral logic. A major insight is the necessity of shifting from lane-based path planning to area-based navigation, as traffic in these regions operates on drivership, social, and empirical expectations rather than strict legal norms. Consequently, our study demonstrates that synthetic data generation is not only a cost-saving tool, but also an essential required for the future of training automobiles for driving assistance to safely model rare, edge cases and indigenous vehicle classes (like Desi Nosimons and Rickshaws) that are absent from global datasets yet vital for local deployment.

Chapter 2

Literature Review

2.1 Preliminaries

The central challenge in training autonomous vehicles is Reality Gap. It is the difference between the simulation environment where an AV agent is trained and the physical real world where it must operate. As noted in recent surveys (Song et al., 2024; Hu et al., 2023), this gap manifests in two primary forms: One is Visual Gap, which are the differences in texture, lighting and weather rendering between computer graphics and real camera feeds. Another one is Physical Gap, which are the differences in vehicle dynamics, agent behaviour (ex. How simulated pedestrian moves vs the real one) etc.

Simulation Based Learning (SBL) is a training methodology where intelligent models learn through a virtual environment rather than interacting with the real world. The learning loop looks something like this:

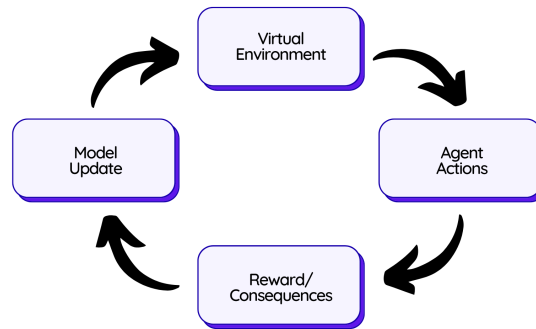


Figure 2.1: Simulation Based Learning

To address the reality gap, three distinct simulation paradigms have emerged. They are Sim2Real, Digital Twins, and PI (Parallel Intelligence). Let's look at Sim2Real, which refers to the transfer of knowledge learned from a virtual environment to the real world. It uses techniques like Domain Randomization and Domain Adaptation, where we deliberately vary simulation parameters and try to find the gap between simulated and real data to learn. Digital Twins is a virtual replica of the real world that can be updated in real-time. Parallel Intelligence is a framework where artificial

systems (simulated) and neural systems (real-world) operate in a parallel manner while improving each other.

Synthetic data generation primarily utilizes three approaches: traditional Computer Graphics (CG) for physical precision, Game Engine-based simulations (e.g., CARLA) for creating dynamic, on-demand "corner cases," and Generative AI (GANs) for photorealistic style transfer and scenario creation (Song et al., 2024; Leudet et al., 2018).

The concept of Drivership has been proposed to characterize good driving behaviour (Fraade-Blanar et al., 2024). This framework works on defining an agent's behavior in a simulated environment, it could be both individual and collective expectations of a simulation.

Standard datasets represent structured and western traffic environments but in south asian environments, these datasets fail due to unique class imbalances and chaotic behavior. The presence of Heterogenous Traffic (ex. Non-standard vehicles like Rickshaw, CNGs) propose a unique challenge.

2.2 Review of Existing Research

Complexities are part of training Autonomous Vehicles to achieve fully safe driving on the road. First of all, it is costly and dangerous to run them without being fully prepared. According to the research, billions of miles should be driven by AVs to reach the reliability of human drivers [13]. As a consequence, simulation has arrived as a core necessity from a helpful side tool. The problem of collecting massive amounts of training data, usually too expensive to collect, is solved only because of the synthetic data generation process [23]. Synthetic datasets allow engineers to increase control over generating high-quality training scenarios. They can accurately adjust lighting, set weather conditions like rain or fog, and create complex traffic scenes that are to be found in real life [23]. The importance of this capability is proven because standard testing misses long-tail events—rare but critical safety scenarios that might happen only once every 30,000 miles [13], [23].

Beyond just volume, diversity in training data is crucial to prevent models from failing in new environments, especially to prevent overfitting. Standard datasets generally face difficulties from a "sunny weather bias" or geographic limitations, primarily featuring well structured roads in the USA or Germany [8], [19]. Synthetic data bridges this gap by providing scope to the researchers to generate diverse environmental conditions like heavy monsoon rain, extreme glare, or night-time chaos—ensuring that AV prediction systems work correctly in all situations [8], [17]. This diversity is more crucial in the South Asian subcontinent, specifically countries like Bangladesh or India. Since collecting real-world data for every such scenario like indigenous vehicles with unique visual shapes, ignoring strict lane discipline, and various behaviour by the pedestrians is difficult, simulation offers a way to digitally recreate these unstructured environments and indigenous vehicles [24]. This allows AVs to be trained safely for the chaotic reality of cities like Dhaka or Delhi before hitting the physical roads [26], [28].

To make the simulations useful and effective, we must fix the reality gap, which refers to the difference between the virtual world and the real world. Recent stud-

ies categorize the methods to do this into three main categories: Sim2Real, Digital Twins, and Parallel Intelligence [13]. To maintain the safety of the validation of the sensors and algorithms, these virtual testbeds are crucial. Because of them, researchers get the scope to test how a car handles dangerous failures or extreme driving limits without anyone putting himself at actual risk on public roads [4], [5]. As we mentioned, we have found that a significant "domain gap" exists in the unstructured and mixed traffic of South Asia. Most current technologies perform well on structured highways by focusing on predictable backgrounds and clearly marked lanes [6]. However, this gap represents not just a technical obstacle, but a fundamental shift in how a vehicle must perceive and interact with its surroundings and other road users. This gap indicates the lackings in the datasets for traffic and road scenarios for any other countries except the USA and Germany [8], [19].

The first layer of this gap is visual. As established by Baig et al. [16] in the BadODD dataset, Bangladeshi roads are filled with unique vehicle classes, such as human-powered rickshaws, CNGs, and modified three-wheelers that are completely missing from the "alphabet of traffic" defined by Western datasets. A similar scenario appears in the IDD dataset [6], which highlights that the diversity within vehicle classes on Indian roads is significantly higher. For example, a "truck" in Delhi may look entirely different from a "truck" in Berlin, causing standard detection models to fail.

The structural gap is even more profound. Modern autonomous systems are "lane-dependent," relying on geometric rules to plan paths. However, the traffic in South Asia is "area-based." As highlighted in the survey by Min et al. [18], lane markings are load-bearing infrastructure for every major path-planning system in use today, and the moment those markings are treated as optional, where [24] demonstrates through UAV-collected data that lanes are often treated as theoretical suggestions rather than physical boundaries. Instead of following fixed lanes, vehicles, particularly smaller ones like motorcycles and rickshaws, navigate a "continuous drivable surface." This behavior makes standard prediction models ineffective and requires a shift toward "free-space" estimation, where navigation is decided by the physical size of the available gap rather than the legality of the lane.

Finally, the gap extends to human behavior. While the PIE dataset [7] successfully uses AI to predict a pedestrian's intent to cross a street in Toronto, these models are blind to the social cues of a South Asian street. In chaotic traffic, "good driving" or Drivership [25] is defined by mutual understanding and the non-verbal behaviors of pedestrians. Furthermore, as the ARAI India-Specific Scenario report [28] concludes, unless we recreate specific regional hazards like "cattle on highways" or "wrong-way driving"—in our virtual environments, our autonomous systems will remain technically functional but practically useless in the region.

Establishing a robust autonomous driving framework needs understanding what "success" is. The literature suggests that we have two benchmarks for it, one is technical verification and another one is social validation. Technical validation ensures sensor reliability and social validation measures the ability of an agent to integrate into human flow of traffic. The baseline of technical performance is well established in structured environments. [5] conducted a performance analysis of

level 2 autonomous systems using MATLAB route modelling. This confirms that under ideal conditions, such as clear weather and standardized lane marks, there is no notable difference between automated systems and manual human driving. This validates the current generation of sensor hardware like LiDAR, radar, and camera fusion that are mechanically capable of safe navigation. However, [5] explicitly noted that the stability is based on “good road infrastructure, which is a critical dependency. It shows that while the hardware is ready, the software reliance on infrastructure will be a point of failure for the unstructured roads of south asian roads. As we move on to the chaotic, “area-based”, heterogeneous traffic of South Asia, technical metrics like “lane deviation” become irrelevant because lanes do not exist. [25] argue that evaluation must shift from pure safety to “Drivership”, the ability of an agent to align with the mutual expectations of other agents in the road.

Western models are trained almost exclusively on Normative Expectations (rules, norms, laws). However, South Asian traffic operates heavily on Empirical Expectations (what will likely happen based on experience). Fraade-Blanar et al. conclude that for an AV to be considered “good” in a specific locality, it must satisfy the empirical expectations of that region [25]. We are not just using synthetic data to teach the car to “see” rickshaws (Technical), but to teach it the “Empirical” behaviors (Social) required to navigate among them without causing severe issues.

The final challenge in our methodology is the effective addition of synthetic scenes into modern E2E (End to End) autonomous driving architectures. While synthetic generation allows us to create the “chaotic” paths of heterogeneous traffic, feeding the data into models trained on real world sensors presents a “Domain Gap”.

The adaptation of CNN made pattern recognition in computer vision revolutionary, and so has been for autonomous vehicles. Features are automatically learnt from its training data, which makes training a model faster. Implementation of CNN algorithms on parallel computing graphics processing units made automobile driving assistance seem like a possibility of tomorrow. There are levels of autonomous driving vehicles, ranging from level 1 to 5. Up to level 3, human attention to driving is necessary, and driving intervention may be required anytime [2].

Standard E2E models such as UniAd and VAD are sensor-dependent which need raw camera or LiDAR inputs to work. In the context of our research, generating photorealistic sensor data for diverse and unique vehicle classes is computationally prohibitive and prone to visual artifacts. Kim et al. [27] identifies this bottleneck, understanding that relying solely on real-world sensor data limits the variety of training scenarios, leaving the system vulnerable to rare, unrecorded events. To solve this issue, Kim et al. [27] introduces SynAD, a framework designed to enhance real world models using synthetic data without complex sensor rendering. The core innovation is the Map-to-BEV network, which acts as a bridge between simulation and perceptions. In this network, SynAD projects path level scenarios directly into high dimensional BEV (Bird’s Eye View) feature space [11]. This effectively allows the model to learn from spatial logic of the scenario without being distracted by the visual fidelity of the simulation, and is suited for “Area-Based” navigation challenges like the domain gap we found in our literature review.

2.3 Summary of Key Findings

Paper / Author	Methodologies	Findings	Year
End to end learning for self-driving cars (Bojarski et al.)	Trained a CNN model to map raw camera data directly to steering wheel.	CNN learned entire task of lane and road following without manual decomposition into road or lane marking detection, semantic abstraction, path planning, and control.	2016
Virtual environment for training autonomous vehicles (Leudet et al.)	Virtual simulation environments (Unity/Game engines) mimicking real sensors (Lidar/Camera).	Confirmed simulation as a dual-purpose tool for safely testing algorithms in dangerous scenarios and generating training data.	2018
Performance analysis of vehicle-specific methods... (Pucher et al.)	Comparison of manual vs. semi-autonomous driving using MATLAB/Simulink and vehicle dynamics sensors.	AVs perform comparably to humans in good infrastructure; sensor reliability drops significantly in adverse weather/road conditions.	2018
IDD: A dataset for exploring problems... (Varma et al.)	IDD (Indian Driving Dataset); 4-level label hierarchy for unstructured traffic.	Western models (e.g., trained on Cityscapes) fail in India; identified unique classes like "drivable area" vs. road.	2018
PIE: A large-scale dataset... (Rasouli et al.)	PIE Dataset; RNN/LSTM models for pedestrian intention prediction.	Context (traffic lights, looking at car) is crucial; intention prediction requires temporal analysis, not just static pose.	2019
BDD100K: A diverse driving dataset... (Yu et al.)	BDD100K (100k videos); Heterogeneous Multitask Learning.	Validated that high diversity in weather, time, and location is essential to prevent model overfitting; enables multi-task learning.	2020

Paper / Author	Methodologies	Findings	Year
Bevformer: Learning bird's-eye-view representation... (Li et al.)	BEVFormer (Spatiotemporal Transformers); nuScenes.	Achieved state-of-the-art 3D detection and velocity estimation by transforming multi-camera inputs into a unified Bird's-Eye-View.	2022
How simulation helps autonomous driving... (Hu et al.)	Survey of Sim2Real, Digital Twins, and Parallel Intelligence.	Simulation is evolving from simple testing to closed-loop "Parallel Intelligence" systems to bridge the Reality Gap.	2023
BadODD: Bangladeshi autonomous driving object detection dataset (Baig et al.)	BadODD dataset; Object detection for regional vehicles.	Established a dataset containing unique Bangladeshi vehicles (Rickshaws, CNGs) absent in global datasets.	2024
Improving end-to-end autonomous driving... (Goel & Narasimhan)	SynDiff-AD (Latent Diffusion Models); Synthetic variations of standard sets.	Improved segmentation (Mask2Former) by +1.4 mIoU and planning by 20% by synthesizing rare weather conditions (rain/fog).	2024
Autonomous driving in unstructured environments... (Li et al.)	Survey of path planning in unstructured traffic.	Highlighted that "lane-based" planning fails in unstructured roads; proposed shift to "area-based" or "free-space" navigation.	2024
A survey on autonomous driving datasets... (Liu et al.)	Statistical analysis of 265 datasets; "Impact Score".	Identified severe geographic bias (45% US/Germany); confirmed lack of "long-tail" safety scenarios in naturalistic data.	2024
Synthetic datasets for autonomous driving: A survey (Song et al.)	Survey of synthetic generation techniques (Game engines vs. Generative AI).	Synthetic data acts as a "data factory" for corner cases where real-world data collection is too expensive or dangerous.	2024

Paper / Author	Methodologies	Findings	Year
Microscopic vehicle trajectory datasets... (Ali et al.)	UAV (Drone) video collection in India; Trajectory data.	Vehicles in mixed traffic ignore lanes and utilize lateral space ("creeping"); defined non-linear flow dynamics of heterogeneous traffic.	2025
Being good (at driving): Characterizing behavioral expectations... (Fraade-Blanar et al.)	Conceptual framework (Drivership).	Differentiated between Normative (legal) and Empirical (social) expectations; AVs must learn social behaviors to be accepted.	2025
Evaluating YOLO architectures... (Hossain et al.)	YOLO v8, v9, v10 evaluation on Bangladesh traffic footage.	Generic YOLO models fail on indigenous vehicles (Leguna, Nosimon); requires architecture-specific tuning for local classes.	2025
SynAD: Enhancing real-world end-to-end... (Kim et al.)	SynAD framework; Map-to-BEV network.	Enhances real-world models by projecting synthetic path-level scenarios into BEV space, avoiding visual domain gap artifacts.	2025
Developing India-specific virtual scenarios... (Pachhapurkar)	Real-to-Sim workflow; IPG CarMaker; Scenario Builder.	Created a library of validated "cut-in" and "cattle on road" scenarios specifically for testing ADAS on Indian roads.	2025

Table 2.1: Key Findings and Summary

Chapter 3

Proposed Methodology

3.1 Methodology Overview

Our proposed methodology develops a region-specific synthetic data generation framework for producing realistic adverse driving scenarios, with the aim of pre-training autonomous vehicle detectors on conditions that real-world collection cannot easily provide. The central challenge it addresses is the persistent scarcity of annotated data representing South Asian traffic — particularly scenarios involving indigenous vehicle classes, unusual traffic behaviours, and hazardous road events. Four guiding constraints shape the entire design: visual realism must be maintained, the sim-to-real gap minimised, annotations produced automatically, and the dangers of collecting genuinely hazardous footage on real roads avoided. The pipeline operates in two complementary modes — a still-image generation pipeline (Mode A) and a temporal scenario generation pipeline (Mode B). Across both modes, Stable Diffusion XL (SDXL) [12], [21] is the sole generator of pixels; it operates in exactly two places: the two-pass inpaint in Mode A, and the generation of start and end keyframes in Mode B. Everything else in the pipeline exists to serve SDXL or to check its output. The prompt, mask, and three control images are all preparation. The verification, labelling, and quality filtering stages are all downstream. ControlNet does not generate anything independently; it feeds structural constraints into SDXL’s denoising process so that what gets painted respects the original scene geometry. Wan is a separate model entirely, brought in only for the motion generation step in Mode B. The methodology begins with real driving backgrounds drawn from BadODD and supplementary dashcam footage. These backgrounds supply authentic road geometry, illumination conditions, infrastructure characteristics, and existing environmental context, forming the structural foundation of every generated frame. To introduce region-specific concepts that SDXL’s pretrained weights do not contain, Low-Rank Adaptation (LoRA)[10] modules are trained using small reference image sets corresponding to indigenous We only train a LoRA where the base model genuinely falls short — each locally rare vehicle class that SDXL cannot render convincingly gets its own adapter, and anything the base model already handles well is left alone. Before generation, scene understanding is applied to each background using two pretrained networks. SegFormer reads the background and produces a colour-coded map of what each region is — road surface, vehicle, sky, sidewalk, and so on. Depth Anything produces a matching per-pixel depth estimate, giving us the spatial distances and perspective foreshortening we need to place agents convinc-

ingly. Crucially, these three outputs the segmentation map, the depth map, and the edge map serve as control images for ControlNet[15]; they constrain the structure of the generated content but do not contribute to the text prompt. Behavioural scenarios are defined through template-based prompt generation. Scenario templates specify the target event, the participating agents, and their associated behavioural states; each behavioural state is translated into concrete prompt words that directly determine what SDXL paints. Simultaneously, response zones are computed through geometric reasoning to identify plausible placement regions within the scene. These zones have two roles: they determine where the binary mask is drawn, and they inform the wording of the second-pass prompt but they are never passed as numeric coordinates to SDXL. Context-aware binary masks are then constructed from the zone geometry to define exactly which regions of the image may be modified during inpainting. The prepared prompts, masks, and control images are supplied to a Multi-ControlNet[15] conditioned SDXL inpainting framework. ControlNet acts as an adapter — feeding the structural maps into SDXL’s UNet — rather than as an independent generator. In Mode A, a two-pass inpainting strategy is employed. The first pass synthesises the primary agent within the masked region, repainting only that area while leaving all other real pixels untouched. The second pass then receives the first pass’s output as its input image; the mask is repositioned over the surrounding traffic area, and the previously placed agent — now sitting beneath a black, preserved region of the mask — remains completely intact while SDXL reshapes the nearby traffic behaviour to match the scenario. The Canny edge map, depth map, and segmentation map are all derived from the original background and computed once; they are reused in both passes without modification. In Mode B, the Mode A pipeline is executed twice on the same background to produce start and end keyframes that differ only in the agent’s behavioural state — the start frame captures the agent before the event and the end frame captures the resolved state. Wan 2.2 First-Last-Frame-to-Video (FLF2V))[29] then generates the intermediate frames connecting the two endpoints. Unlike classical frame interpolation methods, which blend pixels between two frames and introduce ghosting on real motion, Wan is a video diffusion model that synthesises physically plausible in-between content — generating motion states that neither keyframe contains. The resulting clip captures the agent across many positions, scales, and poses, yielding dozens of varied labelled frames from a single keyframe pair. Generated outputs are verified automatically using Grounding DINO[20], which performs open-vocabulary detection to confirm the presence of intended objects. For video sequences, Grounding DINO seeds detection on frame zero and Grounded SAM 2 then propagates that segmentation mask forward and backward through every frame in the clip, providing consistent object boundaries at each timestep without re-detecting per frame. For still images, SAM refines the detected boundaries by tracing the agent’s actual pixel outline rather than relying on the rough placement box used during scene planning; the final annotation box is fitted around that outline, ensuring labels reflect where the agent genuinely is rather than where it was intended to be. After the annotation pass, a set of simple geometric rules screens each generated frame: the inserted agent cannot physically occupy the same space as a nearby vehicle, a clear gap must exist around it, its base must sit on the road rather than floating above it, and the traffic state visible in the surrounding scene must be consistent with whatever the agent is supposed to be doing. Frames that fail any check are discarded before they can

pollute the annotation pool. What passes gets quality-filtered using FID, KID, and FVD, and whatever clears those thresholds is merged with real-world data for training and evaluation. When too many outputs fail verification, or when evaluation numbers on target classes are not improving, we go back and adjust the prompts, zone placements, mask boundaries, or ControlNet weights, then run the whole thing again. The loop closes when the data quality reaches the level we need. The complete methodology is illustrated in the proposed pipeline shown in Figure 3.1.

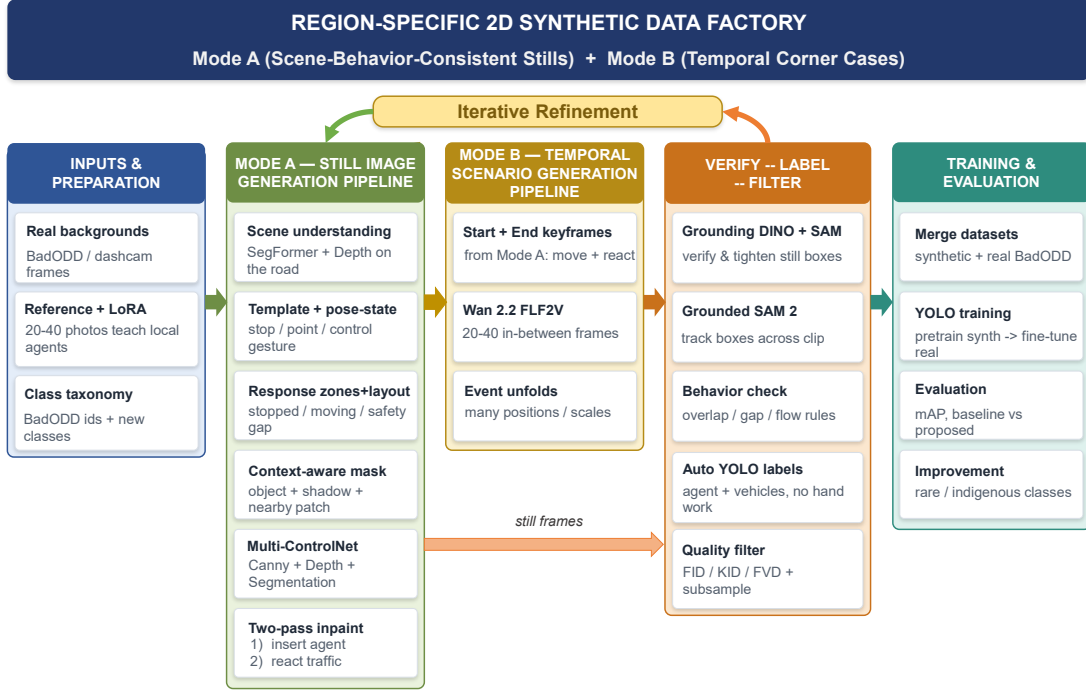


Figure 3.1: Methodology

3.2 Preliminary Design Specification

Several design alternatives were evaluated before arriving at the proposed framework. Traditional simulation environments such as CARLA provide physically grounded scenarios and automatic annotations; however, they require significant effort to create assets representative of South Asian traffic the indigenous vehicles, road markings, and infrastructure that define the target domain. Moreover, the visual appearance of simulated environments typically differs substantially from real dashcam imagery, compounding the sim-to-real problem rather than reducing it. Generative text-to-image pipelines were also investigated. While fully generative pipelines are able to create a variety of scenarios, the creation of an entire frame from scratch leads to unrealistic road geometry, inconsistency of spatial layout, and inability to maintain precise object relations issues becoming even more critical given the spatial nature of the traffic intersection. In order to solve the problems inherent in both sets of methods, a hybrid generation pipeline was designed. Instead of creating something from scratch, we began with actual photographs and then edited those photographs. All the frames in our pipeline begin as photographs taken by a

dashcam; the role of diffusion-based inpainting is limited to adding a new agent or event at the specified location. We have selected SDXL for our generation purposes due to its superiority in quality and ControlNet compatibility. ControlNet adapters receive such inputs as Canny edges, depth maps, and segmentation maps, all of which are extracted from the source photograph. This helps us to stick to actual scene geometry rather than generating something fake-looking. LoRA modules address the indigenous concept gap: each rare class that the base SDXL model fails to represent adequately gets its own targeted adapter trained on 20–40 reference images, while common vehicle types SDXL already knows need none. The two-pass inpainting strategy was chosen over single-pass generation because it decouples agent placement from environment adjustment and the inserted agent is locked in place in the first pass while the surrounding traffic is independently modified in the second. For temporal generation, Wan 2.2 FLF2V was chosen over classical frame interpolation methods such as RIFE or FILM. Classical approaches work by blending pixels between two endpoint frames and introduce ghosting wherever significant real motion is involved. Wan, by contrast, is a video diffusion model that generates physically plausible intermediate content, producing motion states that neither keyframe contains, such as a limb mid-gesture or a wheel mid-turn which is what makes the resulting sequences look genuinely natural rather than blended. The final design therefore consists of the following components:

- Real-world driving backgrounds,
- Scenario-specific prompt generation,
- LoRA-based indigenous concept adaptation,
- SegFormer[9] semantic segmentation,
- Monocular depth estimation,
- Context-aware mask generation,
- Multi-ControlNet conditioned SDXL inpainting,
- Wan FLF2V video generation,
- Grounding DINO[20] and SAM verification,
- Automatic YOLO annotation generation.

Each of those choices addresses a specific failure mode in the alternatives we considered together they produce a pipeline that is realistic enough to train on, controllable enough to target specific rare classes, and scalable enough to produce a dataset of meaningful size without manual annotation effort.

3.3 Data Collection

Our dataset is not purely synthetic but real imagery underpins everything. The actual road surfaces, vehicle distributions, and lighting conditions in every generated frame come from genuine dashcam footage, with BadODD[22] as the primary source. For LoRA training, small reference image sets are gathered for each indigenous vehicle category and rare traffic agent in scope. Twenty to forty representative images are typically sufficient to adapt SDXL to a given local concept. Because each rare class that the base SDXL model cannot handle requires its own LoRA while common classes need none, the collection effort is concentrated precisely on the gap classes rather than spread across the whole taxonomy. The synthetic component of the dataset is generated automatically by the diffusion-based pipeline. Scenario templates are paired with diverse real backgrounds to produce a wide range of adverse driving conditions that would be either too dangerous or logistically impractical to capture from real roads.

3.3.1 Data Cleaning

Real-world datasets undergo a cleaning pass before use. Duplicate images, severely corrupted files, and visually unusable samples are removed from the dataset. For the reference sets used in LoRA training, images with excessive occlusion, poor visibility, or incorrect class representation are excluded to ensure the adapter learns the intended concept cleanly. After synthetic generation, an additional quality control pass is applied. Grounding DINO verification identifies cases where intended objects were not successfully produced, and those samples are automatically discarded. Geometry-based behaviour checks further eliminate unrealistic outputs involving object overlap, floating agents, or inconsistent traffic responses.

3.3.2 Data Transformation

Several transformation steps occur at different points in the pipeline. During scene understanding, semantic segmentation maps and depth maps are extracted from each original background. Canny edge detection[1] is also applied to obtain structural edge representations of the scene. Prompt transformation converts predefined behavioural states into textual descriptions appropriate for SDXL conditioning. Context-aware masks are converted into binary representations marking editable and non-editable regions. For detector training, verified object annotations are expressed in YOLO-compatible format, consisting of normalised centre coordinates and bounding box dimensions.

3.3.3 Data Integration

The final training dataset is assembled by combining multiple sources. Real-world images from BadODD are combined with synthetic still images generated through Mode A. Temporal sequences generated through Mode B are incorporated by extracting verified frames at predefined intervals. Annotations from Grounding DINO and SAM on the synthetic side are merged directly with the existing real-world labels from BadODD to form a single YOLO-format training split. The detector sees

both authentic road conditions and the rare synthetic scenarios in the same batch neither source alone would cover the full range of what we need it to learn.

3.3.4 Data Reduction

If left unchecked, the Mode B sequences would flood the dataset with near-identical frames. We prevent that by subsampling at fixed intervals, picking one representative frame every N timesteps rather than keeping everything Wan produces. The control images act as an implicit reduction on top of that segmentation maps, depth representations, and Canny edges carry only the structural content that matters for generation, stripping out irrelevant variation. FID[3], KID, and FVD scores then provide a distributional checkpoint: any sample that sits too far from the real-world statistics gets dropped before training begins.

3.3.5 Summary of Preprocessed Data

What comes out the other end is a YOLO-format dataset[14] where every sample is an RGB image paired with a text file of class labels and normalised bounding boxes. The class set covers standard driving categories alongside the local classes that motivated this project: rickshaws, auto-rickshaws, and the other indigenous South Asian vehicle types that standard benchmarks have never included. The synthetic portion is what gives the dataset its behavioral range dangerous close-contact scenarios and adverse driving conditions that genuine dashcam footage would rarely capture by chance are all present, but they sit within the visual domain of real Bangladeshi roads rather than looking like they came from a rendering engine.

3.4 Implementation of Selected Design

Implementation begins by selecting a real driving background from the prepared dataset. SegFormer and Depth Anything are applied to produce a semantic segmentation map and a depth map respectively, while Canny edge detection is run on the original background to extract a structural edge representation of the scene. All three control images the segmentation map, the depth map, and the Canny edge map are derived from the original background and computed once; they are reused in both inpainting passes without any modification. Concurrently, scenario templates generate textual prompts describing the intended traffic event. Response zones are defined through geometric reasoning to identify plausible locations for object insertion. These zones determine both where the binary mask is drawn and how the second-pass prompt is worded. Binary masks for the identified zones are subsequently generated using OpenCV. The prompt, mask, segmentation map, depth map, and Canny edge map are supplied to a Multi-ControlNet conditioned SDXL inpainting framework. During the first inpainting pass, the primary agent is synthesised within the selected region; SDXL repaints only the masked area while all other real pixels remain untouched. In the second pass, the output of the first pass becomes the new input image, and the mask is repositioned over the surrounding traffic area rather than the agent. The previously placed agent now falls beneath a black, preserved region of the mask and is kept completely intact while SDXL adjusts the nearby traffic behaviour to match the scenario. For temporal scenarios,

the still-image pipeline is executed twice on the same background to produce start and end keyframes. The two keyframes differ only in the agent’s behavioural state: the start frame captures the agent before the event and the end frame captures the resolved state. Wan 2.2 FLF2V receives the two keyframes along with a short motion prompt and generates the intermediate frames connecting them. Unlike classical frame interpolation, which merely blends pixels and introduces ghosting on real motion, Wan synthesises genuinely plausible in-between content rendering motion states that neither endpoint frame contains. The resulting clip provides the agent at many positions, scales, and poses across the sequence, yielding dozens of varied labelled frames from a single keyframe pair. Following generation, Grounding DINO performs open-vocabulary detection to verify the presence of intended objects. For video sequences, detection is seeded on frame zero and Grounded SAM 2 then propagates the resulting segmentation mask through every frame of the clip, supplying consistent object boundaries at each timestep without re-detecting per frame. For still images, the rough placement box used during inpainting planning is not necessarily where the agent ended up SDXL paints within the masked region at its own scale and exact position. SAM corrects for this by tracing the actual pixel boundary of the agent; the annotation box is then drawn around that trace, so the label reflects where the agent genuinely is rather than where the mask was drawn. Finally, a number of geometric constraints are enforced on the output that is no spatial overlap with other vehicles, a definite safe distance around the agent, ground contact verification, and matching traffic situation according to the behavior of the agent. Those frames which fail one constraint or another are removed at once, and the successful examples are automatically annotated into YOLO format. Finally, distribution metrics (FID, KID, and FVD) are calculated, and those which stray too far from the real distribution are removed from the set. After this stage, the dataset is mixed with BadODD and trained. Our method of training comes in two phases. The first trains only with synthetic data, allowing the model to learn the difficult indigenous classes and adverse situations without ever seeing an actual image. The second phase is used to fine-tune using actual BadODD video data. When outputs from the verification pass or the quality metrics fall short too many rejections, or mAP on target classes not moving we revisit the generation settings, adjust, and run again. Final performance is measured in mAP and per-class AP, with the indigenous vehicle categories given particular weight since those are the classes the whole project exists to address.

Chapter 4

Conclusion

The findings of our project on Synthetic Data Generation of Adverse Driving Conditions for Autonomous Driving Pretraining provide an important framework for overcoming the safety challenge in autonomous driving training for the diverse traffic conditions in South Asia. Through our synthetic data creation and scalability efforts, we wish to successfully identify vehicle types like “Desi Nosimons” and auto-rickshaws and simulate dangerous driving scenarios based on areas which are not present in conventional Western datasets. Additionally, through our experiment, we were able to show that synthetic data was not merely a replacement for actual data collection efforts but were essential in order to cover all the important edge cases in order for the autonomous cars to be prepared for transitioning from test tracks to reality. Looking towards the future, our focus would lie in evaluating our pre-trained models against real world benchmarks, namely through the use of BadODD for quantifying the reduction in the “Sim-to-Real” Gap. Lastly, our goal will be to improve the realism of our simulation to an extent that synthetic data will lead to the development of autonomous vehicles with enhanced safety and cultural compatibility in the Global South.

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